Delhi Public School

Subject: Physics | Grade: 12th | Class: 12th

Time: 1 Hour | Maximum Marks: 30

Name: ________________________ Roll No: ___________ Section: ___________

General Instructions:

1. *Read all questions carefully before answering.*
2. *Write neatly and legibly.*

Section A

Answer all questions in this section. Questions 1 to 5 are Multiple Choice Questions, questions 6 to 10 are True/False questions, and questions 11 to 15 are Fill in the Blanks. Each question carries 1 mark.

Q1. Two point charges placed at a distance 'r' in air exert a force 'F' on each other. If the distance between them is doubled, the new force acting between them will be: [Easy] **[1]**

a. F/2
b. F/4
c. 2F
d. 4F

Q2. If the potential difference applied across a given copper conductor is doubled, the drift velocity of free electrons inside the conductor will: [Moderate] **[1]**

a. remain unchanged
b. be halved
c. be doubled
d. become four times

Q3. Lenz's law of electromagnetic induction is a direct consequence of which of the following conservation laws? [Moderate] **[1]**

a. Conservation of charge
b. Conservation of mass
c. Conservation of energy
d. Conservation of linear momentum

Q4. In a series LCR alternating current circuit, at resonance, the phase difference between the current and the source voltage is: [Hard] **[1]**

a. 0 radians
b. $\pi/2$ radians
c. $\pi/4$ radians
d. π radians

Q5. The de Broglie wavelength of an electron accelerated through a potential difference of V volts is given by the expression: [Moderate] **[1]**

a. $1.227 / \sqrt{V}$ nm
b. $12.27 / \sqrt{V}$ nm
c. $0.1227 / \sqrt{V}$ nm
d. $122.7 / \sqrt{V}$ nm

Q6. The electrostatic potential inside a charged hollow spherical conductor is zero. [Easy] **[1]**

Q7. A magnetic dipole placed in a uniform magnetic field always experiences a net non-zero translational force. [Moderate] **[1]**

Q8. Self-inductance of a coil is also referred to as the electromagnetic inertia of the electrical circuit. [Easy] **[1]**

Q9. When a light wave travels from an optically denser medium to an optically rarer medium, its frequency decreases. [Moderate] **[1]**

Q10. The binding energy per nucleon is maximum for the iron nucleus (Fe-56), making it highly stable. [Hard] **[1]**

Q11. The SI unit of electric flux is ________. [Easy] **[1]**

Q12. The magnetic susceptibility of a diamagnetic substance is always ________. [Moderate] **[1]**

Q13. A ________ is an electrical device used to step-up or step-down the alternating voltage based on mutual induction. [Easy] **[1]**

Q14. The focal length of a plane mirror is ________. [Moderate] **[1]**

Q15. In a p-n junction diode, the width of the depletion region ________ when a reverse bias voltage is applied. [Hard] **[1]**

Section B

Answer all short answer questions. Each question carries 2 marks.

Q16. State Gauss's Law in electrostatics and write down its mathematical expression for a charge enclosed in a closed surface. [Easy] **[2]**

Q17. A wire of resistance R is stretched uniformly such that its length is doubled. Calculate the new resistance of the wire in terms of its original resistance. [Moderate] **[2]**

Q18. Explain why the iron core of a transformer is laminated. What specific energy loss does this design minimize? [Hard] **[2]**

Section C

Answer all long answer questions. Each question carries 3 marks.

Q19. Derive an expression for the electric field intensity at a point on the axial line of an electric dipole of dipole moment 'p'. [Moderate] **[3]**

Q20. State the Biot-Savart Law in vector form. Use this law to derive the expression for the magnetic field at the center of a circular current-carrying loop of radius R. [Moderate] **[3]**

Q21. With the help of a neat circuit diagram, explain the working of a junction diode as a full-wave rectifier. Sketch the input and output waveforms. [Hard] **[3]**

— End of Question Paper —